

On the Treewidth and Pathwidth of Biconvex Bipartite Graphs

Sheng-Lung Peng* and Yi-Chuan Yang

Department of Computer Science and Information Engineering
National Dong Hwa University
Hualien 97401, Taiwan
slpeng@mail.ndhu.edu.tw

Abstract. In this paper we explore the biclique structure of a biconvex bipartite graph G . We define two concatenation operators on bicliques of G . According to these operations, we show that G can be decomposed into two chain graphs G_L and G_R , and a bipartite permutation graph G_P . Using this representation, we propose linear-time algorithms for the treewidth and pathwidth problems on biconvex bipartite graphs.

1 Introduction

Let $G = (V, E)$ be a finite, simple, and undirected graph where V and E denote the vertex and edge sets of G , respectively. G is called *bipartite* if V can be partitioned into two sets X and Y such that every edge has its ends in different sets. Equivalently, G is bipartite if and only if it is 2-colorable and is denoted as $G = (X, Y, E)$. A *biclique* is a complete bipartite subgraph. Let $N(v) = \{w \mid (v, w) \in E\}$ and $N[v] = N(v) \cup \{v\}$.

A graph G is a *permutation graph* if there is some pair P, Q of permutations of the vertex set such that there is an edge between vertices u and v if and only if u precedes v in one of P, Q , while v precedes u in the other. *Bipartite permutation graphs* simultaneously have both of the properties of bipartite graphs and permutation graphs [20].

A bipartite graph $G = (X, Y, E)$ is *biconvex* if both X and Y can be ordered so that for every vertex v in $X \cup Y$ neighbors of v occur consecutively in the ordering [1]. We say that X and Y each possess the convexity property, and that $V = X \cup Y$ has a biconvex ordering.

A graph is *chordal* if it does not contain a chordless cycle of length greater than three. A *triangulation* of a graph $G = (V, E)$ is a graph $H = (V, E')$ such that H is a supergraph of G and H is chordal. The *treewidth* of graph G is the minimum k such that there is a triangulation of G with maximum clique size at most $k + 1$ [3]. Given a graph G and an integer k , the treewidth problem asks whether the treewidth of G is at most k . This problem remains NP-complete on cobipartite graphs (complement of bipartite graphs) [2], bipartite graphs [13], and graphs of maximum degree at most 9 [8]. For some special classes

* Corresponding author.

of graphs, it has been shown that it can be computed in polynomial time, such as cographs [7], circular-arc graphs [21], chordal bipartite graphs [15], permutation graphs [5,18], circle graphs [12], d -trapezoid graphs [6], and distance hereditary graphs [9].

A graph $G = (V, E)$ is an *interval graph* if one can put the vertices into one-to-one correspondence with a set of intervals $\{I_u\}_{u \in V}$ on the real line, such that $(u, v) \in E$ if and only if $I_u \cap I_v \neq \emptyset$. The *pathwidth* of any graph G is the minimum k such that G can be embedded into an interval graph with maximum clique size at most $k + 1$. Given a graph G and an integer k , the pathwidth problem asks whether the pathwidth of G is at most k . The pathwidth problem is NP-complete on cobipartite graphs [2], bipartite graphs [13], chordal graphs [10], cocomparability graphs [11], chordal domino graphs [16], bipartite distance-hereditary graphs [14], and planar graphs [19]. The known graph classes for which the pathwidth can be computed in polynomial time are cographs [7], permutation graphs [5,18], d -trapezoid graphs [6], split graphs [10], and graphs with bounded treewidth [4].

In this paper, we show that a bipartite permutation graph can be represented by concatenating a sequence of bicliques. Then we extend this representation to a biconvex bipartite graph. By using this representation we propose linear-time algorithms for the treewidth and pathwidth problems on biconvex bipartite graphs. The best algorithm for solving the treewidth problem on biconvex bipartite graphs runs in $O(m^3)$ time where m is the number of edges in the graph [15]. However, the complexity of the pathwidth problem on biconvex bipartite graphs is unknown. The only known result is that it is NP-complete on chordal bipartite graphs (implied from [14]).

2 Bipartite Permutation Graphs

2.1 Biclique Structure

Let $G = (X, Y, E)$ be a bipartite graph. An ordering of the vertices in X has the *adjacency property* if for each vertex y in Y , vertices of $N(y)$ are consecutive in the ordering of X . An ordering of the vertices in X has the *enclosure property* if for every pair of vertices v, w in Y such that $N(v)$ is a subset of $N(w)$, vertices in $N(w) \setminus N(v)$ occur consecutively in the ordering of X . A *strong ordering* of the vertices of $G = (X, Y, E)$ consists of an ordering of X and an ordering of Y such that for all edges $(x, y), (x', y')$ in E where x, x' are in X and y, y' are in Y , $x < x'$ and $y > y'$ imply (x, y') and (x', y) are in E . The edges (x, y) and (x', y') are called a *crossing pair*. Spinrad *et al.* give the following characterization of bipartite permutation graphs.

Theorem 1. [20] The following statements are equivalent.

1. $G = (X, Y, E)$ is a bipartite permutation graph.
2. There is a strong ordering of $X \cup Y$.
3. There exists an ordering of X which has the adjacency and enclosure properties.